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Systems and methods for network-based slice access authorization

Abstract

A method may include receiving, by at least one network device and from a user device, a registration message including a service identifier and at least one of a network slice identifier or a network slice token. The method may also include determining, based on information included in the registration message, whether the user device is authorized to use a network slice associated with the service identifier. The method may further include setting up a data session to be serviced by the network slice, in response to determining that the user device is authorized to use the network slice.

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Background/Summary

BACKGROUND INFORMATION

(1) The concept of network slices within Fifth Generation (5G) networks is one of the benefits of 5G. Slices may be associated with a particular Quality of Service (QoS). User devices typically need to be authorized to use network slices to avoid overloading the network slices with unauthorized data traffic. Despite authorization requirements, however, network slices are still vulnerable to unauthorized usage due to the security of the user device execution environment being compromised, fraud or other reasons.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an exemplary environment in which systems and methods described herein may be implemented;
- (2) FIG. 2 is a block diagram of components implemented in the environment of FIG. 1 in accordance with an exemplary implementation;
- (3) FIG. 3 illustrates logic components implemented in one or more of the devices illustrated in FIGS. 1 and 2 in accordance with an exemplary implementation;
- (4) FIG. 4 is a flow diagram illustrating processing associated with network slice access authorization in accordance with an exemplary implementation;
- (5) FIG. 5 is an exemplary signal flow diagram associated with the processing of FIG. 4;
- (6) FIG. 6 illustrates an exemplary database used in connection with the processing of FIG. 4; and
- (7) FIG. 7 is another flow diagram illustrating exemplary processing associated with network slice access authorization in accordance with another exemplary implementation.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

(8) The following detailed description refers to the accompanying drawings. The same reference

numbers in different drawings may identify the same or similar elements.

(9) Implementations described herein provide for network slice access authorization to avoid unauthorized network slice usage. In exemplary implementations, network elements in a core network associated with routing data traffic provide a user device initiated solution to authorizing network slice usage. For example, in one implementation, a user device requests and receives network slice information, which is later verified by the core network prior to setting up a data session serviced by the requested network slice. In another implementation, an application server may initiate the network slice authorization process by providing slice related information to a user device. The network slice authorization information may then be verified by core network elements prior to setting up a data session serviced by the requested slice. In each case, core network elements determine whether the user device is authorized to use a network slice to provide a unified approach to network slice authorization. In this manner, unauthorized network slice usage may be avoided.

(10) FIG. 1 is a diagram illustrating an exemplary environment **100** in which systems and methods described herein may be implemented. Referring to FIG. 1, environment **100** includes user equipment (UE) device **110-1** through **110-N**, access network **120**, wireless stations **122-1** through **122-N**, core network **130**, network devices **140**, data network **150** and application service provider **160**.

(11) UE devices **110-1** and **110-2** (referred to herein individually as UE device or UE **110**, and collectively as UE devices or UEs **110**) may include any computing device, such as a personal computer (PC), a laptop computer, a server, a tablet computer, a notebook, a Chromebook®, a mobile device, such as wireless or cellular telephone device (e.g., a conventional cell phone with data processing capabilities), a smart phone, a personal digital assistant (PDA) that can include a radiotelephone, any type of mobile computer device or system, a game playing device, a music playing device, a home appliance device, a home monitoring device, a virtualized system, etc., that includes communication functionality. UE device **110-1** may connect to access network **120** via wireless station **122-1** and UE device **110-N** may connect to access network **120** via wireless station **122-N**. UE devices **110** may also connect to other devices in environment **100** via any conventional technique, such as wired, wireless, optical connections or a combination of these techniques. UE device **110** and the person associated with UE device **110** (e.g., the party holding or using UE device **110**) may be referred to collectively as UE device **110** or UE **110** in the description below.

(12) Access network **120** may provide access to transport network **130** for wireless devices, such as UE devices **110**. Access network **120** may enable UE device **110** to connect to core network **130** for Internet access, non-Internet Protocol (IP) data delivery, cloud computing, mobile telephone service, Short Message Service (SMS) message service, Multimedia Message Service (MMS) message service, and/or other types of data services. Access network **120** may provide access to core network **130**, a service or application layer network, a cloud network, a multi-access edge computing (MEC) network, a fog network, etc. Furthermore, access network **120** may enable a device in core network **130** to exchange data with UE device **110** using a non-IP data delivery method such as Data over Non-Access Stratum (DoNAS).

(13) Access network **120** may also include a 5G access network or another advanced network, such as a Fourth Generation (4G) Long Term Evolution (LTE) access network. Additionally, access network **120** may include functionality such as the functionality of a millimeter (mm) Wave Radio Access Network (RAN). Access network **120** may also include: support for advanced or massive multiple-input and multiple-output (MIMO) antenna configurations (e.g., an 8×8 antenna configuration, a 16×16 antenna configuration, a 256×256 antenna configuration, etc.); support for cooperative MIMO (CO-MIMO) configurations; support for carrier aggregation; relay stations; Heterogeneous Networks (HetNets) of overlapping small cells and macrocells; Self-Organizing Network (SON) functionality; machine type communication (MTC) functionality, such as 1.4 MHz

wide enhanced MTC (eMTC) channels (also referred to as category Cat-M1), Low Power Wide Area (LPWA) technology such as Narrow Band (NB) IoT (NB-IoT) technology, and/or other types of MTC technology; and/or other types of 5G functionality.

(14) Wireless stations **122** (referred to collectively as wireless stations **122** and individually as wireless station **122**) may be included in access network **120**. Each wireless station **122** may service a number of UE devices **110** and/or other user devices when the particular device is within radio frequency range of wireless station **122**. In one implementation, wireless station **122** may include 5G base station (e.g., a next generation NodeB (gNB)) that includes one or more radio frequency (RF) transceivers. For example, wireless station **122** may include three RF transceivers and each RF transceiver may service a 120 degree sector of a 360 degree field of view. Each RF transceiver may include or be coupled to an antenna array. The antenna array may include an array of controllable antenna elements configured to send and receive 5G new radio (NR) wireless signals via one or more antenna beams. In other implementations, wireless station **122** may also include a 4G base station (e.g., an evolved NodeB (eNodeB)) or a 6G base station that communicates wirelessly with UEs **110** located within the radio frequency range of wireless station **122**.

(15) Core network **130** may include one or more wired, wireless and/or optical networks that are capable of receiving and transmitting data, voice and/or video signals. In an exemplary implementation, core network **130** may be associated with a telecommunications service provider (e.g., a service provider providing cellular wireless communication services and wired communication services) and may manage communication sessions of UE devices **110** connecting to core network **130** via access network **120**. Core network **130** may include one or multiple networks of different types and technologies. For example, core network **130** may be implemented to include a next generation core (NGC) network for a 5G network, an Evolved Packet Core (EPC) of an LTE or LTE Advanced network, a sixth generation (6G) network, and/or a legacy core network. Core network **130** may provide packet-switched services and wireless IP connectivity to various components in environment **100**, such as UE devices **110**, to provide, for example, data, voice, and/or multimedia services.

(16) Core network **130** may include various network devices **140**. Depending on the implementation, network devices **140** may include 5G core network components (e.g., a User Plane Function (UPF), an Access and Mobility Management Function (AMF), a Network Exposure Function (NEF), a Session Management Function (SMF), a Unified Data Management (UDM) function, a Unified Data Repository (UDR), a Policy Control Function (PCF), a Charging Function (CHF), etc.), 4G core network components (e.g., a Serving Gateway (SGW), a Packet data network Gateway (PGW), a Mobility Management Entity (MME), a Home Subscriber Server (HSS), a Policy Charging and Rules Function (PCRF) etc.), or another type of core network components (e.g., future 6G network components). In other implementation, network devices **140** may include combined 4G and 5G functionality, such as a session management function with PGW-control plane (SMF+PGW-C) and a user plane function with PGW-user plane (UPF+PGW-U).

(17) Data network **150** may include, for example, a packet data network. In an exemplary implementation, UE device **110** may connect to data network **150** via core network **130**. Data network **150** may also include and/or be connected to a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), an autonomous system (AS) on the Internet, an optical network, a cable television network, a satellite network, a wireless network, an ad hoc network, a telephone network (e.g., the Public Switched Telephone Network (PSTN) or a cellular network), an intranet, or a combination of networks.

(18) Application service provider **160** may include one or more computer devices and/or servers that store applications (referred to herein as “apps”) and provide the applications and services associated with the applications to users, such as UE devices **110**. For example, application service provider **160** may provide a particular application accessed by UE device **110** via core network

130. In some implementations, application service provider **160** may also provide network slice tokens to UE devices **110**, as described in detail below.

(19) The exemplary configuration illustrated in FIG. **1** is provided for simplicity. It should be understood that a typical environment may include more or fewer devices than illustrated in FIG. **1**. For example, environment **100** may include a large number (e.g., thousands or more) of UE devices **110** and wireless stations **122**, as well as multiple access networks **120** that connect UE devices **110** to core network **130**. Environment **100** may also include elements, such as gateways, monitoring devices, network elements/functions, etc. (not shown), that aid in providing data services and routing data in environment **100**.

(20) Various functions are described below as being performed by particular components in environment **100**. In other implementations, various functions described as being performed by one device may be performed by another device or multiple other devices, and/or various functions described as being performed by multiple devices may be combined and performed by a single device.

(21) FIG. **2** illustrates a portion of environment **100**, including elements implemented in core network **130** in accordance with an exemplary implementation. Referring to FIG. **2**, network devices **140** in core network **130** include AMF **142**, SMF **143**, UPF **144**, NSSF **145**, NEF **146**, UDM/UDR **147** and PCF **148**. It should be understood that core network **130** may include other elements and/or differently arranged elements. Environment **100** also includes application repository **210**, operational support system (OSS) **220** and authorization device **230**.

(22) As illustrated in FIG. **2**, UE device **110-1** may connect to core network **130** via wireless station **122**, shown in FIG. **2** as gNB **122**. Application service provider **160** may also connect to elements in core network **130**, such as UPF **144** and NEF **146**.

(23) Application repository **210** may store applications accessed and provided to UE devices **110**. For example, application repository **210** may correspond to an “app store” via which users access applications. In some implementations, application repository **210** may store applications provided by application service provider **160**. In an exemplary implementation, application repository **210** may store an application identifier (ID) and a corresponding service ID for each application. OSS **220** may provide services associated with applications stored in application repository **210**. For example, in one implementation, OSS **210** may generate application IDs and service IDs for each application in application repository **210** and provide this information for storage in application repository **210**.

(24) Authorization device **230** may include a computing device or server that generates tokens or other authorization information for use in authorizing access to a network slice, as described in detail below. For example, in one implementation, authorization device **230** may correspond to an OAuth 2.0 authorization server that generates tokens for use by user devices **110** executing particular applications to access network slices during data sessions. Authorization device **230** and/or other devices in, for example, core network **130** may validate the token using, for example, a public key or other authentication mechanism to determine whether the token is valid.

(25) AMF **142** may perform registration management, connection management, reachability management, mobility management, lawful intercepts, Short Message Service (SMS) transport, transport of session management messages between UE device **110** and other network devices, such as SMF **143**, access authentication and authorization, location services management, functionality to support non-3GPP access networks, and/or other types of management processes. In an exemplary implementation, AMF **142** may perform slice access authorization, as described in detail below.

(26) UPF **144** may maintain an anchor point for intra/inter-RAT mobility, maintain an external Packet Data Unit (PDU) point of interconnect to a particular data network (e.g., data network **150**), perform packet routing and forwarding, perform the user plane part of policy rule enforcement, perform packet inspection, perform lawful intercept, perform traffic usage reporting, perform QoS

handling in the user plane, perform uplink traffic verification, perform transport level packet marking, perform downlink packet buffering, forward an “end marker” to a RAN node (e.g., gNB **122**), and/or perform other types of user plane processes.

(27) SMF **143** may perform session establishment, session modification, and/or session release, perform IP address allocation and management, perform Dynamic Host Configuration Protocol (DHCP) functions, perform selection and control of UPF **144**, configure traffic steering at UPF **144** to guide the traffic to the correct destinations, terminate interfaces toward PCF **148**, perform lawful intercepts, charge data collection, support charging interfaces, control and coordinate of charging data collection, terminate session management parts of Non-Access Stratum (NAS) messages, perform downlink data notification, manage roaming functionality, and/or perform other types of control plane processes for managing user plane data.

(28) NSSF **145** may select a set of network slice instances to serve a particular UE device **110**, determine network slice selection assistance information (NSSAI) or a Single-NSSAI (S-NSSA), determine a particular AMF **142** to serve a particular UE device **110**, and/or perform other types of processing associated with network slice selection or management.

(29) NEF **146** may expose capabilities and events to other network functions (NFs), including third party NFs, application functions (AFs), edge computing NFs, and/or other types of NFs.

Furthermore, NEF **146** may secure provisioning of information from external applications to core network **130**, translate information between core network **130** and devices/networks external to core network **130**, support a Packet Flow Description (PFD) function, and/or perform other types of network exposure functions. In an exemplary implementation, NEF **146** may store information regarding network slice access and perform network slice access authorization checking, as described in detail below.

(30) UDM/UDR **147** may maintain subscription information for UE devices **110**, manage subscriptions, generate authentication credentials, handle user identification, perform access authorization based on subscription data, perform network function registration management, maintain service and/or session continuity by maintaining assignment of SMF **143** for ongoing sessions, support SMS delivery, support lawful intercept functionality, and/or perform other processes associated with managing user data.

(31) PCF **148** may support policies to control network behavior, provide policy rules to control plane functions (e.g., to SMF **143**), access subscription information relevant to policy decisions, render policy decisions, and/or perform other types of processes associated with policy enforcement.

(32) Environment **100** illustrated in FIG. 2 may include additional elements and/or NFs that are not illustrated. It should also be understood that functions described as being performed by various elements in FIG. 2, including elements in core network **130**, may be performed by other elements/functions in other implementations.

(33) FIG. 3 illustrates an exemplary configuration of a device **300**. One or more devices **300** may correspond to or be included in devices in environment **100**, such as UE device **110**, wireless station **122**, network devices **140**, such as AMF **142**, SMF **143**, UPF **144**, NSSF **145**, NEF **146**, UDM/UDR **147**, PCF **148**, application service provider **160**, application repository **210**, OSS **220**, authorization device **230** and other devices included in environment **100**. Referring to FIG. 3, device **300** may include bus **310**, processor **320**, memory **330**, input device **340**, output device **350** and communication interface **360**. The exemplary configuration illustrated in FIG. 3 is provided for simplicity. It should be understood that device **300** may include more or fewer components than illustrated in FIG. 3. Processor **320** may include one or more processors, microprocessors, or processing logic that may interpret and execute instructions. Memory **330** may include a random access memory (RAM) or another type of dynamic storage device that may store information and instructions for execution by processor **320**. Memory **330** may also include a read only memory (ROM) device or another type of static storage device that may store static information and

instructions for use by processor **320**. Memory **330** may further include a solid state drive (SSD). Memory **330** may also include a magnetic and/or optical recording medium (e.g., a hard disk) and its corresponding drive.

(34) Input device **340** may include a mechanism that permits a user to input information, such as a keypad, a keyboard, a mouse, a pen, a microphone, a touch screen, voice recognition and/or biometric mechanisms, etc. Output device **350** may include a mechanism that outputs information to the user, including a display (e.g., a liquid crystal display (LCD)), a speaker, etc. In some implementations, device **300** may include a touch screen display may act as both an input device **240** and an output device **350**.

(35) Communication interface **360** may include one or more transceivers that device **300** uses to communicate with other devices via wired, wireless or optical mechanisms. For example, communication interface **360** may include one or more radio frequency (RF) transmitters, receivers and/or transceivers and one or more antennas for transmitting and receiving RF data.

Communication interface **360** may also include a modem or an Ethernet interface to a LAN or other mechanisms for communicating with elements in a network.

(36) In an exemplary implementation, device **300** performs operations in response to processor **320** executing sequences of instructions contained in a computer-readable medium, such as memory **330**. A computer-readable medium may be defined as a physical or logical memory device. The software instructions may be read into memory **330** from another computer-readable medium (e.g., a hard disk drive (HDD), solid state drive (SSD), etc.), or from another device via communication interface **360**. Alternatively, hard-wired circuitry may be used in place of or in combination with software instructions to implement processes consistent with the implementations described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

(37) FIG. **4** is a flow diagram illustrating processing associated with network slice access authorization in accordance with an exemplary implementation. FIG. **5** is a signal flow diagram illustrating exemplary signal flows associated with the processing of FIG. **4**. Processing may begin with an application being stored in application repository **210**. For example, OSS **220** or another device in environment **100** may approve an application for storage in application repository **210**. OSS **220** may assign an application identifier (ID) to the application and a service ID to the application (block **410**). OSS **220** may assign the service ID to applications stored in app repository **210** to effectively tie particular classes of applications to particular service categories. The service categories may then be assigned to network slices. When a UE **110** accesses applications stored in app repository **210**, the UE **110** receives the application ID and service ID and uses this information for network slice access authorization, as described in detail below.

(38) For example, OSS **220** may assign a particular service ID to gaming applications, assign another service ID to industrial Internet of Things (IIoT) applications, such as an application associated with ultra reliable low latency communications (URLLC), assign still another service ID to business related applications, etc. In this case, assume that the application is a video game. OSS **220** may assign an application ID, such as Game X to the application, and a Service ID, such as Game Network Slice (NS)-1 to the application, indicating that Game X can be serviced with NS **1**. OSS **220** may forward the application ID and service ID to NEF **146** (block **410**; FIG. **5**, **510**).

(39) NEF **146** receives the application information for Game X and registers the application by storing the application ID and service ID in a database within NEF **146** (block **420**; FIG. **5**, **520**). For example, FIG. **6** illustrates an exemplary database **600** included in NEF **146**. In alternative implementations, database **600** may be located externally from NEF **146** (e.g., in AMF **142**, SMF **143**, NSSF **145**, etc.) and accessible by NEF **146**, AMF **142**, SMF **143**, NSSF **145**, etc. Referring to FIG. **6**, database **600** includes a UE ID field **610**, a Network Slice ID field **612**, a Service ID field **614**, an Application ID field **616**, a Session ID field **618** and a token field **620**.

(40) As illustrated in row **600-1** of table **600**, NEF **146** may store a service ID of GameNS-1 in

field **614** and an application ID of Game X in field **616**. NEF **146** may also access NSSF **145** to obtain a network slice ID, such as network slice selection assistance information (NSSAI), that identifies a particular slice for servicing a session associated with UE **110-1** executing Game X. In this example, NSSF **145** may identify network slice selection assistance information (NSSAI) that uniquely identifies a particular slice, such as NSSAI-1. NSSF **145** may forward this information to NEF **146** for storage in database **600** (e.g., in NS ID field **612** in row **600-1**, as illustrated in FIG. **6**). NEF **146** may also optionally store an access token and/or information identifying an access token for NSSAI-1, such as information identifying token 1 (T1) in token field **620** of row **600-1**, as also illustrated in FIG. **6** (block **420**). The token, if used, provides additional security with respect to ensuring a UE **110** is authorized to use a particular network slice, as described in detail below.

(41) As also illustrated in row **600-2** of table **600**, a NS ID, service ID may be stored for another application, accessed by another UE **110**. For example, a UE **110** identified with UE ID **999** may be associated with an IIOT application Y (stored in app ID field **616** of row **600-2**) and having a service ID of URLLC Network Slice (stored in service ID field **614**). As illustrated, different applications have different service IDs, NS IDs, use different tokens, etc.

(42) Assume that a user at UE **110-1** wishes to play Game X. In an exemplary implementation, UE **110-1** may access Game X from application repository **210** or from storage within UE **110-1** if Game X has already been downloaded by the user at UE **110-1**. In either case, when Game X is executed/launched, UE **110-1** may transmit a request for a network slice access token to NEF **146** (FIG. **5**, **530**). NEF **146** receives the request for the slice access token (block **430**). NEF **146** may then identify the particular UE **110-1** based on a UE ID included in the request, the application ID (i.e., Game X) and/or the service ID (GameNS-1) and obtain the access token associated with Game X and the corresponding service ID (block **532**).

(43) For example, NEF **146** may access database **600** and identify the access token for Game X in row **600-1**, field **620**. In this example, the access token is T1. As described previously, NEF **146** may obtain network slice selection assistance information (NSSAI) from NSSF **145** and obtain an access token corresponding to the NS ID provided by NSSF **145**. For example, NEF **146** may obtain a slice access token for the particular application from authorization device **230**. In other implementations, the token T1 generated by authorization device **230** may be previously stored in database **600**. In either case, NEF **146** may provide the network slice access token T1 to UE **110-1** (block **430**; FIG. **5**, **540**). UE **110-1** receives the network slice access token.

(44) Assume that UE **110-1** initiates a registration service request associated with a session for playing Game X. For example, UE **110-1** may transmit a registration request message that includes a UE ID associated with UE **110-1**, an Application ID, a Service ID, an NSSAI, and/or the access token received from NEF **146**, as well as other information, such as a data network name (DNN) associated with the request (FIG. **5**, **550**). AMF **142** may receive the registration request message from UE **110-1** (block **440**). AMF **142** may then determine whether UE **110-1** is authorized to access the particular requested network slice based on the UE ID, Service ID, NSSAI and/or token included in the registration message (block **450**).

(45) For example, NEF **146** may determine whether the UE ID, NS ID, Service ID, App ID and/or token provided in the registration message correspond to the information stored in database **600** for the particular UE **110-1** and service ID. For example, AMF **142** may signal NEF **146** to determine whether the UE ID included in the registration message identifies a UE **110** that is authorized to use the particular slice identified by the service ID and NSSAI included in the registration message based on performing a lookup in database **600**. NEF **146** may also signal authorization device **230** to validate the token included in the registration message.

(46) Assume that NEF **146** determines that the information in the registration request message matches the information stored in database **600** and NEF **146**/authorization device **230** determines that the token is valid. In this case, NEF determines that UE **110-1** is authorized to use the

particular network slice identified in the registration message and/or associated with the provided token (block **450**—yes).

(47) NEF **146** may signal AMF **142** that UE **110-1** is authorized to use the network slice (FIG. 5, **570**). AMF **142** and/or NEF **146** may then signal SMF **143** to initiate a PDU session for UE **110-1** with the particular network slice (i.e., NSSAI-1) servicing the PDU session (block **460**; FIG. 5, **580**). SMF **143** may provide PDU session information to UE **110-1**, and SMF **143** may then establish the PDU session using the particular network slice (block **470**; FIG. 5, **590**).

(48) Referring back to block **450**, if AMF **142**/NEF **146** determines that UE **110-1** is not authorized to use the requested network slice for executing Game X (block **450**—no), NEF **146** may signal SMF **143** to initiate a PDU session for UE device **110-1**, but without using the resources corresponding to the particular network slice (block **480**). For example, if the service ID and/or NS ID included in the registration message do not match the information stored in database **600** and/or the access token is not validated, this may indicate that a UE **110** is attempting to access a network slice for which UE is not authorized. In this case, SMF **143** may provide a PDU session for UE **110-1**, but without using resources associated with the requested network slice. In alternative implementations, SMF **143** may not establish a PDU session when UE **110-1** is attempting to access a network slice for which UE **110-1** is not authorized, and SMF **143** may send an alert to, for example, OSS **220**.

(49) In this manner, elements in core network **130** may determine whether a UE **110** is authorized to use a network slice, prior to initiating a PDU session. This may help prevent network slices from becoming overloaded by unauthorized data traffic.

(50) As described above with respect to FIGS. 4 and 5, in one implementation, a UE **110** may initiate a request for network slice authorization and obtain network slice information with respect to particular applications executed by UE **110**. In other implementations, other devices may initiate the provisioning of network slice authorization information to UEs **110**, such as an application service provider/server, as described in detail below.

(51) FIG. 7 is a flow diagram illustrating processing associated with network slice access authorization in accordance with another exemplary implementation. Processing may begin in a similar manner as described above with respect to FIGS. 4 and 5. For example, OSS **220** or another device in environment **100** may approve an application for storage in application repository **210**. OSS **220** may assign an application ID to the application and a service ID to the application (block **710**). The service categories may be assigned to network slices.

(52) For example, OSS **220** may assign a particular service ID to gaming applications, assign another service ID to IIOT applications, assign still another service ID to business related applications, etc. Similar to the implementation described above, assume that the application is a video game. OSS **220** may assign an application ID of Game X to the application, and a Service ID of Game NS-1 to the application, indicating that Game X can be serviced with NS **1**. OSS **220** may forward the application ID and service ID to NEF **146** (block **710**).

(53) NEF **146** receives the application information for Game X and registers the application by storing the application ID and service ID database **600** (block **720**). NEF **146** may also access NSSF **145** to obtain a network slice ID, such as NSSAI that identifies a particular slice for servicing a session associated with UE **110-1** executing Game X, such as NSSAI-1. NEF **146** may obtain a network slice access token for the particular application from authorization device **230**. NEF **146** may also store the access token or information identifying the access token for NSSAI-1, such as information identifying token 1 (T1) in database **600** (block **720**).

(54) In this implementation, NEF **146** may forward the access token and/or information identifying the token to application service provider **160**. Application service provider **160** may then push the token associated with GameNS-1 to UE device **110-1** (block **730**). In an exemplary implementation, NEF **146** may also provide the token to other elements in core network **130**, such as PCF **148**, via, for example, UE route selection policy (URSP) signaling. In each case,

application service provider **160** initiates the process associated with providing a network slice authorization token to UE **110-1**. That is, the token is sent to UE **110-1** without UE **110-1** requesting the token.

(55) Assume that a user at UE **110-1** wishes to play Game X. As described previously, UE **110-1** may access Game X from application repository **210** or from storage within UE **110-1** if Game X has already been downloaded by the user at UE **110-1**. In either case, UE **110-1** may initiate a registration service request associated with a session for playing Game X. For example, UE **110-1** may transmit a registration request message that includes a UE ID associated with UE **110-1**, an Application ID, a Service ID, an NSSAI, and/or the access token received from application service provider **160**, as well as other information, such as DNN associated with the request. AMF **142** may receive the registration request message from UE **110-1** (block **740**). AMF **142** may then determine whether UE **110-1** is authorized to access the particular requested network slice corresponding to the NSSAI and token included in the registration message (block **750**).

(56) For example, as described above with respect to FIGS. **4** and **5**, NEF **146** may determine whether the UE ID, NS ID, Service ID, App ID provided in the registration message correspond to the information stored in database **600** for the particular UE **110-1**. For example, AMF **142** may signal NEF **146** to determine whether the UE identified by the UE ID included in the registration message is authorized to use the particular slice identified by the service ID, NSSAI included in the registration message. NEF **146** may also signal authorization device **230** to validate the token included in the registration message.

(57) Assume that NEF **146** determines that the information in the registration request matches the information stored in database **600** for the particular Service ID and app ID and that the token is valid. In this case, NEF determines that UE **110-1** is authorized to use the particular network slice identified in the registration message and/or associated with the access token (block **750**—yes).

(58) NEF **146** may signal AMF **142** that UE **110-1** is authorized to use the network slice. AMF **142** and/or NEF **146** may then signal SMF **143** to initiate a PDU session for UE **110-1** with the particular network slice (i.e., NSSAI-1) servicing the PDU session (block **760**). SMF **143** may provide PDU session information to UE **110-1** and SMF **143** may then service the PDU session using the particular network slice (block **770**).

(59) Referring back to block **750**, if AMF **142**/NEF **146** determines that UE **110-1** is not authorized to use the requested network slice for executing Game X (block **750**—no), NEF **146** may signal SMF **143** to initiate a PDU session for UE device **110-1**, but without using the resources corresponding to the particular network slice (block **780**). For example, if the NS ID included in the registration message does not match the information stored in database **600** and/or the token is not validated, this may indicate that a UE **110** is attempting to access a network slice for which UE **110** is not authorized. In such cases, SMF **143** may establish the PDU session, but not service the PDU session with the requested network slice. In other implementations, SMF **143** may not set up the PDU session for UE **110-1** when UE **110** is attempting to access a network slice for which UE **110** is not authorized.

(60) Implementations described herein provide for network slice access authorization performed by elements in a core network. For example, implementations described herein provide for network-based network slice authorization prior to setting up a data session. This may allow a service provider to ensure that the use of network slices is limited to authorized users.

(61) The foregoing description of example implementations provides illustration and description, but is not intended to be exhaustive or to limit the embodiments to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of the embodiments.

(62) For example, features have been described above with respect to providing network slice authorization using elements in core network **130**. In other implementations, similar processing may be performed in other portions of environment **100**, such as in a Multi-access Edge Computing

(MEC) platform located, for example, between access network **120** and core network **130**.

(63) Further, while series of acts have been described with respect to FIGS. **4** and **7** and signal flows with respect to FIG. **5**, the order of the acts and signal flows may be different in other implementations. Moreover, non-dependent acts may be implemented in parallel.

(64) It will be apparent that various features described above may be implemented in many different forms of software, firmware, and hardware in the implementations illustrated in the figures. The actual software code or specialized control hardware used to implement the various features is not limiting. Thus, the operation and behavior of the features were described without reference to the specific software code—it being understood that one of ordinary skill in the art would be able to design software and control hardware to implement the various features based on the description herein.

(65) Further, certain portions of the invention may be implemented as “logic” that performs one or more functions. This logic may include hardware, such as one or more processors, microprocessor, application specific integrated circuits, field programmable gate arrays or other processing logic, software, or a combination of hardware and software.

(66) In the preceding specification, various preferred embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

(67) To the extent the aforementioned embodiments collect, store or employ personal information of individuals, it should be understood that such information shall be collected, stored and used in accordance with all applicable laws concerning protection of personal information. Additionally, the collection, storage and use of such information may be subject to consent of the individual to such activity, for example, through well known “opt-in” or “opt-out” processes as may be appropriate for the situation and type of information. Storage and use of personal information may be in an appropriately secure manner reflective of the type of information, for example, through various encryption and anonymization techniques for particularly sensitive information.

(68) No element, act, or instruction used in the description of the present application should be construed as critical or essential to the invention unless explicitly described as such. Also, as used herein, the article “a” is intended to include one or more items. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

Claims

1. A method, comprising: receiving, by at least one network device and from a user device, a registration message including a service identifier, a network slice identifier and a network slice token; determining, based on the network slice identifier and the network slice token, whether the user device is authorized to use a network slice associated with the service identifier; and setting up a data session to be serviced by the network slice, in response to determining that the user device is authorized to use the network slice.
2. The method of claim 1, wherein the at least one network device comprises an access and mobility management function and a network exposure function.
3. The method of claim 1, further comprising: storing, in a database located in a core network and accessible by the at least one network device, an application identifier (ID), a network slice ID, and a network slice token associated with a network slice for each of a plurality of different applications.
4. The method of claim 3, wherein the determining whether the user device is authorized to use the network slice further comprises: accessing the database to determine whether the application

identifier and the network slice ID included in the registration message, for an application executed by the user device, match information stored in the database for the application; and validating the network slice token.

5. The method of claim 1, further comprising: transmitting, to the user device, the network slice token for an application to be executed by the user device.

6. The method of claim 1, wherein the determining comprises determining that the user device is not authorized to use the network slice, the method further comprising: setting up, for the user device, a data session that will not use the network slice.

7. The method of claim 1, further comprising: receiving, by the at least one network device and from the user device, a request for a network slice token, wherein the request includes a service identifier associated with an application; and providing, by the at least one network device, the network slice token based on the service identifier.

8. The method of claim 1, further comprising: providing, to the user device and not in response to a request from the user device, the network slice token.

9. The method of claim 8, wherein the providing is performed by an application server.

10. A system, comprising: at least one device configured to: receive, from a user device, a registration message including a service identifier and a network slice token; determine, based on the service identifier and the network slice token, whether the user device is authorized to use a network slice associated with the service identifier; and set up a data session to be serviced by the network slice, in response to determining that the user device is authorized to use the network slice.

11. The system of claim 10, wherein the at least one device comprises an access and mobility management function and a network exposure function.

12. The system of claim 10, wherein the at least one device is further configured to: store, in a database located in a core network and accessible by the at least one device, an application identifier (ID), a network slice ID, and a network slice token associated with a network slice for each of a plurality of different applications.

13. The system of claim 12, wherein when determining whether the user device is authorized to use the network slice, the at least one device is further configured to: access the database to determine whether the network slice ID and the network slice token included in the registration message, for an application executed by the user device, match information stored in the database for the application; and validate the network slice token.

14. The system of claim 10, wherein the at least one device is further configured to: transmit, to the user device, a network slice token for an application to be executed by the user device.

15. The system of claim 10, wherein when determining, the at least one device is configured to determine that the user device is not authorized to use the network slice, and wherein the at least one device is further configured to: set up, for the user device, a data session that will not use the network slice.

16. The system of claim 10, wherein the at least one device is further configured to: receive, from the user device, a request for a network slice token, wherein the request includes a service identifier associated with an application; and provide, by the at least one device and based on the service identifier, the network slice token.

17. The system of claim 10, wherein the at least one device is further configured to: provide, to the user device and not in response to a request from the user device, the network slice token.

18. The system of claim 17, wherein the at least one device comprises an application sever configured to provide the network slice token.

19. The method of claim 1, further comprising: servicing the data session using the network slice associated with the network slice identifier, in response to determining that the user device is authorized to use the network slice.

20. The system of claim 10, wherein the registration message includes a network slice identifier and the at least one device is further configured to: service the data session using the network slice

associated with the network slice identifier, in response to determining that that user device is authorized to use the network slice.
